# I2C in LPC2148



# Introduction

I2C (Inter Integrated Circuit) is a serial bus interface connection protocol first invented by Philips Semiconductor (Now NXP Semiconductors). It is also called as TWI (two wire interface) since it uses only two wires for communication. I2C uses a handshaking mechanism for communication. Hence, it is also called as acknowledgment based communication protocol.\

I2C works in two modes namely,

- **Master Mode**
  Master is responsible for generating clock and initiating communication

- **Slave Mode**
  Slave receives the clock and responds when addressed by the Master

# I2C interface

Two wires of the I2C interface are SDA (serial data) and SCL (serial clock).

- SDA is Serial Data wire used for data transfer in between master and slave
- SCL is Serial Clock wire used for clock synchronization. Clock is provided by the master

There are two types of data transfer possible on I2C interface depending on Read/Write operation i.e.

Data transfer from Master Transmitter to Slave Receiver and

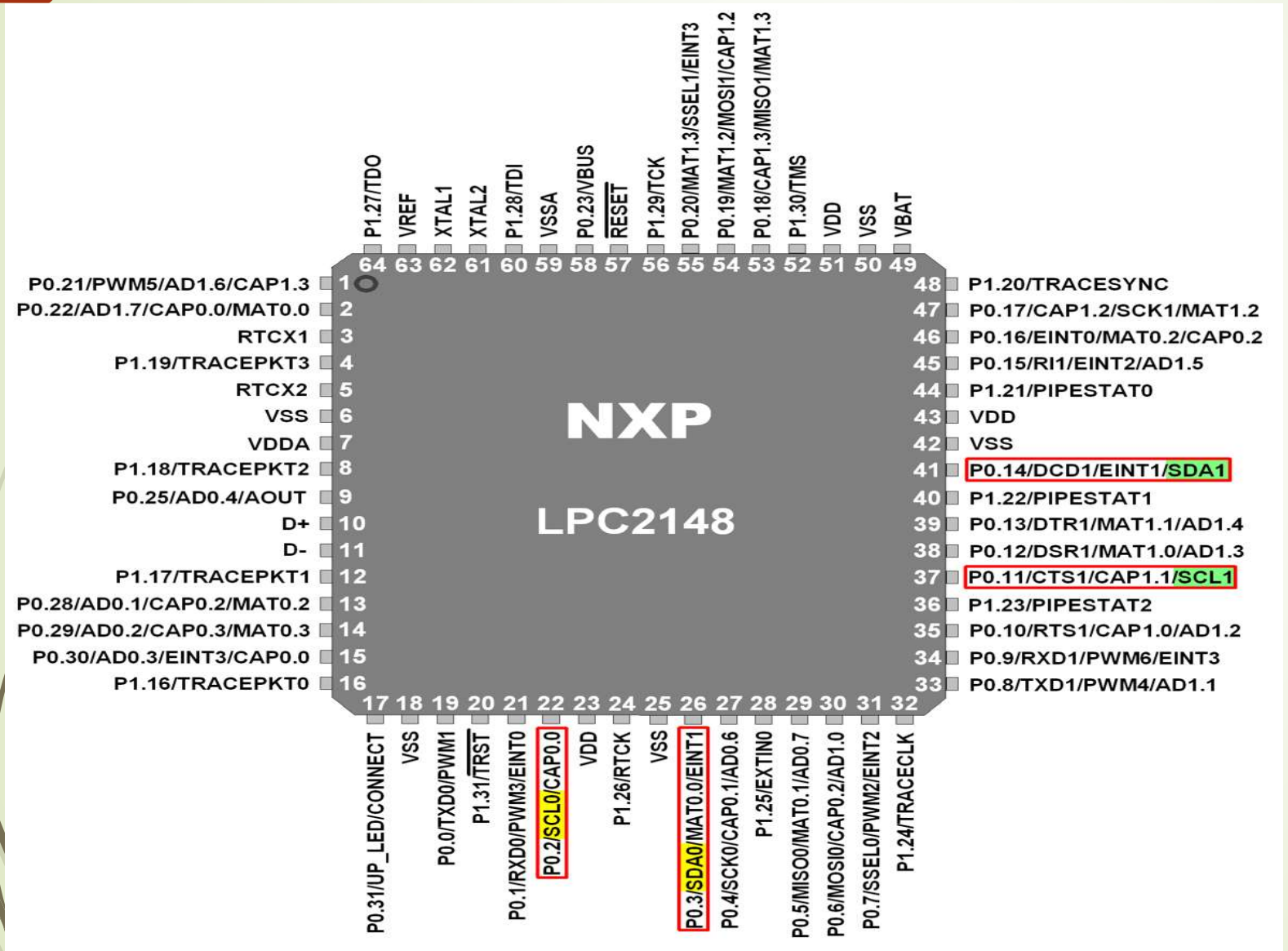
Data transfer from Slave Transmitter to Master Receiver

# LPC2148 I2C

- LPC2148 has two I2C interfaces. i.e. I2C0 & I2C1
- It has Programmable clock to support adjustable I2C transfer rates
- LPC2148 I2C is byte oriented and has four operating modes:

    1. Master Transmitter mode.
    2. Master Receiver mode.
    3. Slave Transmitter mode.
    4. Slave Receiver mode.

# ARM LPC2148 I2C Pins




**SCL0: -** Serial Clock pin of I2C0.

**SDA0: -** Serial Data pin of I2C0.

**SCL1: -** Serial Clock pin of I2C1.

**SDA1: -** Serial Data pin of I2C1.

Let's see the registers that are used to initialize and control the operation of LPC2148 I2C.

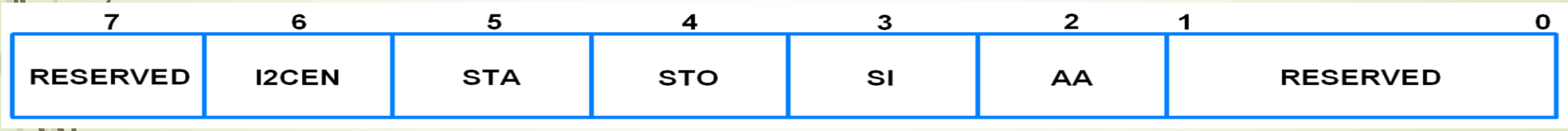
I2C0 and I2C1 have similar register formats and register names. The only difference in their register names is the I2C0 or I2C1 acronym at the beginning of the register name.

Here we will be showing I2C0 registers. Same can be used for I2C1 by changing their name (Replace the I2C0 part by I2C1).

**I2C Registers**

**1. I2C0CONSET (I2C0 Configuration Set Register)**

It is an 8-bit read-write register.
It is used to control the operation of the I2C0 interface.




**I2C0CONSET (I2C0 Configuration Set Register)**

• Writing a 1 to a bit in this register causes corresponding bit in the I2C control register to set.

• Writing a 0 has no effect.

• **Bit 2 – AA (Assert Acknowledge Flag)**
When set to 1, Acknowledge (SDA LOW) is returned during acknowledge clock pulse on SCL. Otherwise, Not acknowledge (SDA HIGH) is returned.

• **Bit 3 – SI (I2C Interrupt Flag)**
This bit is set when the I2C state changes. Else it remains reset.

• **Bit 4 – STO (Stop Flag)**
In Master Mode, setting this bit causes the I2C interface to transmit a STOP condition. In Slave Mode, setting this bit causes recover from an error condition if any.
This bit is cleared by hardware automatically.

• **Bit 5 – STA (Start Flag)**
Setting this bit causes the I2C interface to enter in master mode and transmit a START condition or transmit a repeated START condition if it is already in master mode.

• **Bit 6 – I2CEN (I2C Interface Enable)**
Set this bit to enable I2C interface.

2. **I2C0CONCLR (I2C0 Configuration Clear Register)**

• It is an 8-bit write only register.




**I2C0CONCLR (I2C0 Configuration Clear Register)**

• Writing a 1 to a bit in this register causes corresponding bit in the I2C control register to clear.

• Writing a 0 has no effect.

• **Bit 2 – AAC (Assert Acknowledge Flag Clear)**
Setting this bit clears the AA bit in I2C0CONSET register.

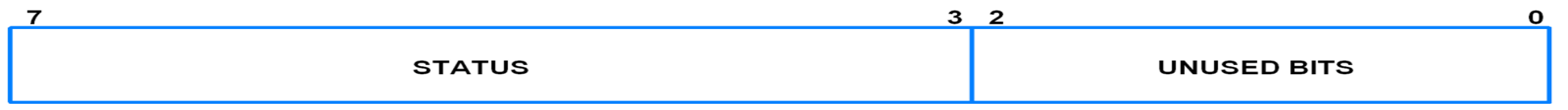
• **Bit 3 – SIC (I2C Interrupt Flag Clear)**
Setting this bit clears the SI bit in I2C0CONSET register.

• **Bit 5 – STAC (Start Flag Clear)**
Setting this bit clears the STA bit in I2C0CONSET register.

• **Bit 6 – I2CENC (I2C Interface Disable)**
Setting this bit clears the I2CEN bit in I2C0CONSET register.

3. **I2C0STAT (I2C0 Status Register)**

- It is an 8-bit read only register.
- It reflects the present status of the I2C interface.




**I2C0STAT (I2C0 Status Register)**

- **Bit 2:0 – Unused bits**

These are unused bits and are always 0.

- **Bit 7:3 – Status**

These are status bits which provide the status of the current event on I2C bus.

There are 26 possible status codes.

**Example,** I2C0STAT = 0x08. This code indicates that a start condition has been transmitted.

For other status codes, refer Table 232 (Page 228) to table 235 of the datasheet given in the attachments section below.

4. **I2C0DAT (I2C0 Data Register)**

• It is an 8-bit read-write register.




**I2C0DAT (I2C0 Data Register)**

• It contains the data to be transmitted or received.

• This register can be read or written to only when SI = 1.

5. **I2C0SCLL (I2C0 SCL Low Duty Cycle Register)**

• It is a 16-bit register.




**I2C0SCLL (I2C0 SCL Low Duty Cycle Register)**

• This register contains the value for SCL Low time of the cycle.

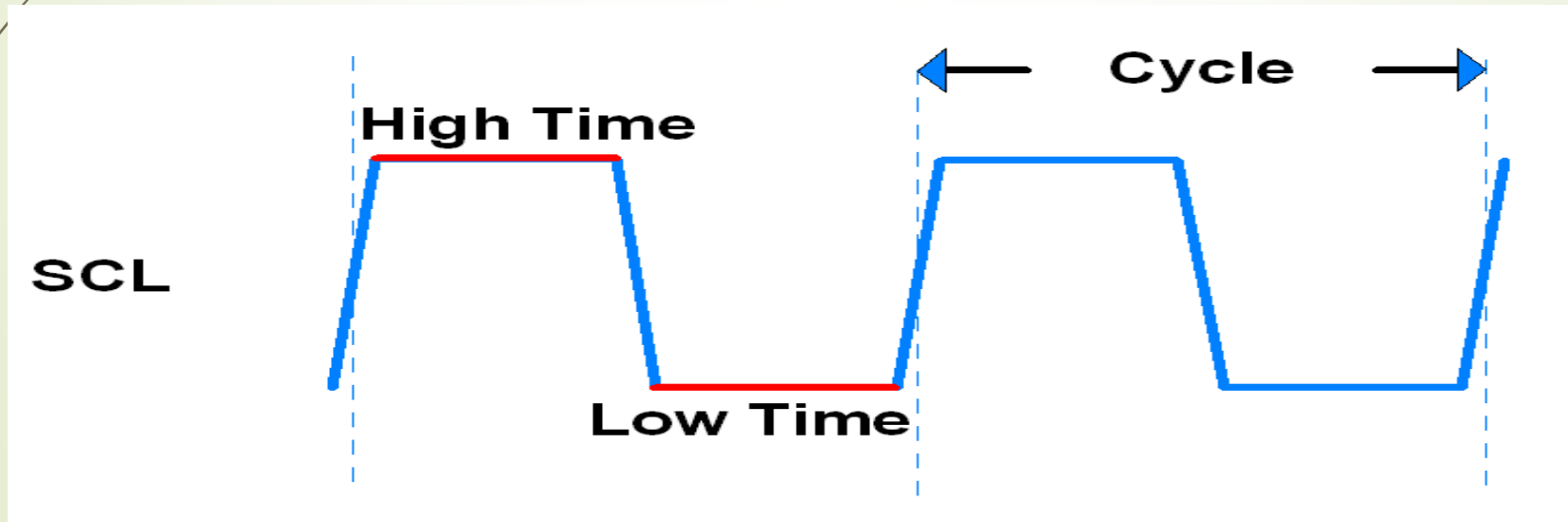
## 6. I2C0SCLH (I2C0 SCL High Duty Cycle Register)

- It is a 16-bit register.




**I2C0SCLH (I2C0 SCL High Duty Cycle Register)**

- This register contains the value for SCL High time of the cycle.



The frequency and duty cycle of SCL is decided using I2C0SCLL and I2C0SCLH. I2C0SCLH contains the TON (High) time and I2C0SCLL contains the TOFF (Low) time.

**The frequency is calculated as follows:**

$$I2CBitFrequency = \frac{Fpclk}{I2C0SCLL + I2C0SCLH}$$

**Example:** Assume FPCLK = 30MHz, I2CBitFrequency = 300KHz, Duty Cycle = 50%.
As duty cycle is 50%, I2C0SCLL = I2C0SCLH.

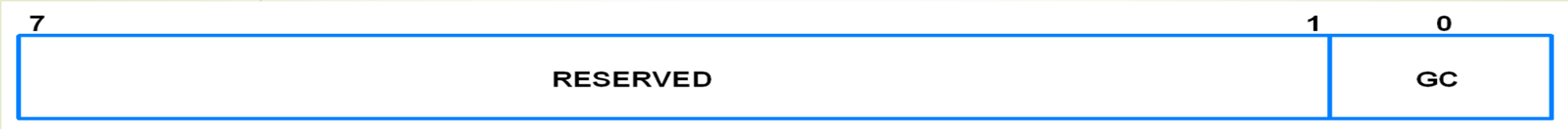
Hence,

$$300000 = \frac{30000000}{2xI2C0SCLH}$$

Hence, I2C0SCLL = I2C0SCLH = 50

## 7. I2C0ADR (I2C0 Slave Address Register)

- It is an 8-bit read-write register.




**I2C0ADR (I2C0 Slave Address Register)**

- It is only used when device is in Slave Mode.

- **Bit 0 – GC (General Call)**
When this bit is set, the general call address (0x00) is recognized.

- **Bit 7:1 – Address**
It contains the I2C device address for slave mode.